

CDI REPORT-UPGRADE

Research Assistant — Required Corrections & Task Directive

Field	Detail
Date	1 May 2026
Task count	8 tasks (2 Critical · 3 Important · 3 Recommended)
Paper target	Q1 Journal — Desalination / Water Research / Sep. Purif. Technol.

1. CURRENT STATE ASSESSMENT

The pipeline is methodologically sound. The core novelty — replacing a fixed charge-efficiency assumption with a learned conductivity proxy Λ_k — is original and publication-worthy. The blocking issue is not the idea; it is the validation implementation and reporting gaps listed below.

Dimension	Status	Impact on Paper
Two-stage GP framework	✓ Sound	Strong basis for Methods section
Λ_k novelty	✓ Original	Core contribution; will anchor Abstract
Ablation study	✓ Good structure	Best current evidence — must be cleaned
Validation loop	✗ Flawed	All reported metrics are invalid
Uncertainty calibration	✗ Overconfident	Pareto bands cannot be reported
Pareto candidate	✗ Incomplete	Cannot be used experimentally
Stage 2 flow handling	⚠ Unjustified	Reviewers will flag this immediately
Publication figures	⚠ Not generated	Required before submission
Scientific language/claims	⚠ Overclaiming	Must be corrected before any drafting

PAPER POSITIONING

This paper should be framed as a methodology contribution, not an optimisation study.
The target framing is: 'A reproducible, cycle-resolved, uncertainty-aware CDI analytics framework that replaces fixed charge-efficiency assumptions with a learned conductivity proxy.'

With only 10 unique operating conditions, claiming optimisation results will invite rejection.
The ablation improvement (~17–21%) is the strongest quantitative claim and must be central.
Target journal: Desalination or Separation and Purification Technology (Q1).

2. TASK PRIORITY MATRIX

ID	Task	Priority	Blocking?
T-01	Fix validation loop (use all cycles)	● CRITICAL	Yes
T-02	Regenerate all results after fix	● CRITICAL	Yes
T-03	Resolve flow variable handling in Stage 2	● IMPORTANT	Yes
T-04	Fix and verify uncertainty calibration	● IMPORTANT	Yes
T-05	Complete Pareto candidate with all coordinates	● IMPORTANT	Yes
T-06	Prepare publication-ready figures (7 total)	● RECOMMENDED	No
T-07	Consolidate and clean ablation table	● RECOMMENDED	No
T-08	Add dataset characterisation + Λ_k framing	● RECOMMENDED	No

3. CRITICAL TASKS

● CRITICAL — Blocking

T-01 Fix the Validation Loop

What: The loop currently saves only the first cycle per held-out condition. All reported MAE/RMSE values are computed from 10 data points instead of ~107. Every published metric is therefore invalid.

Required actions:

- Modify `run_validation()` and `run_ablation()` to iterate over ALL cycles in each held-out condition — not only index [0].
- Output must include: one prediction row per cycle, plus a condition-level summary (mean $\pm$ std) for each held-out condition.
- Confirm corrected output shows `n_cycles` $\approx$ 107 before proceeding to T-02.

Deliverables:

1. Corrected `run_validation()` and `run_ablation()` functions.
2. Validation CSV confirming `n_cycles` $\approx$ 107.
3. Short change log (what was modified and where in the code).

Note: *This is the single highest-priority fix. Nothing else should proceed until this is done.*

T-02 Regenerate All Results

What: Once T-01 is confirmed, rerun the full pipeline. Every number currently in the documentation is based on the flawed loop and must be replaced.

Required actions:

- Rerun Stage 1 (Λ_k), Stage 2 (energy + performance), ablation, and region-based validation.
- Rerun uncertainty calibration with updated LOCO-CV residuals.
- Produce a side-by-side comparison table: old (invalid) vs. corrected metrics.

Deliverables:

4. Updated validation summary (Stage 1 + Stage 2).
5. Updated ablation table (both objectives, all approaches).
6. Updated region-based validation table.
7. Updated calibration table.
8. Old vs. new metric comparison for supervisor sign-off.

4. IMPORTANT TASKS

● **IMPORTANT** — Required for scientific validity

T-03 Resolve Flow Variable Handling in Stage 2

What: Flow was dropped from Stage 2 without documented justification. This creates an incomplete Pareto point and a modeling inconsistency that reviewers will flag. A clear decision is required.

Required actions:

- Option A (preferred): Include flow in Stage 2. This produces a fully defined Pareto candidate (concentration, flow, potential) and removes the inconsistency.
- Option B (acceptable): Keep current structure, but document the ARD sensitivity values for flow in Stage 2, state explicitly what fixed/median flow value was used in the optimisation grid, and prepare a one-paragraph justification for the Methods section.
- Either way: the final Pareto candidate must report all three operating variables.

Deliverables:

9. Written decision: Option A or Option B with supporting sensitivity evidence.
10. Updated Stage 2 model and Pareto result if Option A is chosen.
11. Justification paragraph for the Methods section if Option B is chosen.

Note: Reviewers routinely check whether optimisation results are experimentally reproducible. An incomplete operating point will trigger a revision request.

T-04 Fix Uncertainty Calibration

What: Current nominal coverage: 0% (performance) and 80% (energy) against a 90% target. This is not acceptable for reporting Pareto uncertainty bands in a Q1 paper.

Required actions:

- Diagnose why LOCO-based calibration is not achieving target coverage.
- Correctly apply the empirical multiplier to all three models: Λ_k , energy, performance.
- Verify adjusted coverage falls within 85–95% for all models before using any uncertainty band in a figure.

Deliverables:

12. Calibration table: nominal coverage, empirical multiplier, adjusted coverage — for all three models.
13. Confirmation of 85–95% adjusted coverage.
14. Calibrated uncertainty bands applied to the Pareto front figure.

T-05 Complete the Pareto Candidate

What: The current knee (1000 ppm, 0.947 V) is missing its flow coordinate and has no calibrated uncertainty band. It cannot be reported as-is.

Required actions:

- After T-03 and T-04 are resolved, extract the full knee candidate: (concentration, flow, potential).

- Add calibrated uncertainty bands from T-04 to the predicted R_kappa_pos and SEC_Wh_m3 values.
- Remove all language referring to this point as 'optimal', 'maximum removal', or 'proven optimum'. Replace with: 'model-suggested candidate for experimental validation'.

Deliverables:

15. Complete Pareto knee table: all three operating coordinates + predicted objectives + uncertainty.
16. Updated Pareto front figure with near-observed points, knee marked, calibrated bands shown.

Note: *The paper's contribution is the framework and the methodology, not a specific optimum. Frame the knee as illustrative of how the framework is used, not as its final output.*

5. RECOMMENDED TASKS

RECOMMENDED — Required for publication quality

T-06 Prepare Publication-Ready Figures

What: Seven figures are required for the paper. All must be generated after T-01 through T-05 are complete so they reflect corrected, validated results.

Required actions:

- Fig. 1— Pipeline schematic: raw data → segmentation → cycle metrics → Stage1→ Stage 2 → Pareto.
- Fig. 2 — Cycle segmentation example: current + conductivity vs. time, phase boundaries annotated.
- Fig. 3 — Conductivity removal profile: $\kappa(t)$ for a representative cycle with R_{kappa} area shaded.
- Fig. 4 — $\Lambda\kappa$ sensitivity surface: $\Lambda\kappa$ vs. concentration and potential (flow at median).
- Fig. 5 — Prediction vs. ground truth scatter: one panel each for $\Lambda\kappa$, energy, performance; annotate R^2 and MAE.
- Fig. 6 — Pareto front: SEC vs. $R_{\text{kappa_pos}}$ with calibrated bands, near-observed points, knee marked.
- Fig. 7 — Region-based validation: grouped bar chart comparing near-observed vs. far-extrapolation MAE.
- Export all figures at ≥ 300 DPI (vector PDF or high-resolution PNG).

Deliverables:

17. Seven figures, ≥ 300 DPI, with draft captions.

T-07 Consolidate the Ablation Table

What: The ablation study is the paper's strongest quantitative evidence. It must appear as a single clean table, not scattered across sections.

Required actions:

- After T-02: produce one table — rows = approach (Ridge, GP without $\Lambda\kappa$, GP with $\Lambda\kappa$); columns = Energy MAE/RMSE/Rel. MAE and Performance MAE/RMSE/Rel.MAE.
- Highlight percentage improvement of the full model over each baseline.
- Add a two-sentence interpretive note below the table for the Results section.

Deliverables:

18. Single consolidated ablation table (both objectives, three approaches, improvement % highlighted).
19. Draft interpretive note.

Note: The ~17% energy and ~21% performance improvements are the core quantitative claim of the paper. These numbers must be correct (from T-02) and prominently presented.

What: Two writing tasks that are essential for reviewer credibility: (1) a clear description of the experimental design, and (2) a precise scientific explanation of what Λ_k is and is not.

Required actions:

- Prepare a dataset summary table: sweep type, variable range, conditions per sweep, QC-passing cycles per condition, total cycles.
- Add the following statement (or equivalent) to the documentation: 'This is a sparse experimental design with 10 unique physical conditions and repeated cycle measurements per condition. Results should be interpreted as condition-level trends, not high-density optimisation.'
- Write a Λ_k clarification paragraph: (a) Λ_k is NOT physical charge efficiency; (b) Λ_k IS a conductivity-native proxy — conductivity reduction per unit charge from available sensors; (c) it is valid as a relative comparator within the same system; (d) it cannot be converted to absolute moles removed without Cout calibration.

Deliverables:

20. Dataset characterisation table + one-paragraph description for the Methods section.
21. Λ_k framing paragraph suitable for the Methods or Introduction section.

Note: How Λ_k is positioned relative to existing CDI charge-efficiency literature will determine how reviewers perceive the novelty. The framing paragraph is important for the Introduction.

6. LANGUAGE — REQUIRED CORRECTIONS

The following phrases must be removed from all documentation and any draft text. Overclaiming is one of the most common reasons Q1 reviewers request major revisions.

Remove	Replace With
optimal solution / optimal operating point	model-suggested candidate for experimental validation
maximum removal	highest predicted conductivity removal in near-observed region
true efficiency	conductivity-based efficiency proxy (Λ_k)
proven global optimum	Pareto knee candidate — subject to experimental confirmation
charge efficiency	conductivity-native efficiency proxy